

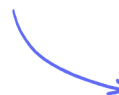
Calculator Questions

Sequences

Introduction to Sequences / nth Terms of Linear Sequences / Quadratic Sequences / Other Sequences

Easy (3 questions)	/7
Medium (3 questions)	/7
Hard (6 questions)	/26
Very Hard (3 questions)	/33
Total Marks	/73

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Easy Questions

- 1 Here is a sequence of numbers.

7, 5, 3, 1, -1, ...

Find the next term in this sequence.

(1 mark)

- 2 Write the next term in each of these sequences.

i) 1 1 2 3 5 8

[1]

ii) 2 4 8 16 32 64

[1]

(2 marks)

- 3 This expression can be used to generate a sequence of numbers.

$$n^2 - n + 11$$

i) Work out the first three terms of this sequence.

[2]

ii) Show that this expression does not only generate prime numbers.

[2]

(4 marks)

Medium Questions

- 1 Find the n th term of the sequence below.

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{1}{10}, \dots$$

(1 mark)

- 2 The n th term of a sequence is $4n^2 + n + 3$.

i) Find the 2nd term.

[1]

ii) Find the value of n when the n th term is 498.

$$n = \dots\dots\dots [3]$$

(4 marks)

- 3 Here is a sequence.

$$a, \quad 13, \quad 9, \quad 3, \quad -5, \quad -15, \quad b, \quad \dots$$

Find the value of a and the value of b .

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(2 marks)

Hard Questions

1 (a) Find the n th term of each sequence.

4 2 0 -2 -4 ...

(2 marks)

(b)

1 7 17 31 49 ...

(2 marks)

2 Complete the table for the 5th term and the n th term of each sequence.

1st term	2nd term	3rd term	4th term	5th term		n th term
9	5	1	-3			
4	9	16	25			
1	8	27	64			
8	16	32	64			

(11 marks)

3 0, 1, 1, 2, 3, 5, 8, 13, 21, ...

This sequence is a Fibonacci sequence. After the first two terms, the rule to find the next term is "add the two previous terms". For example, $5 + 8 = 13$.

Use this rule to complete each of the following Fibonacci sequences.

2	4			
1				11
.....	-1			1

(3 marks)

4 Find the n th term of the sequence below.

1, 5, 25, 125, 625, ...

(2 marks)

5 Find the n th term of each sequence.

i) -1, -3, -5, -7, -9, ...

[2]

ii) 2, 9, 28, 65, 126, ...

[2]

(4 marks)

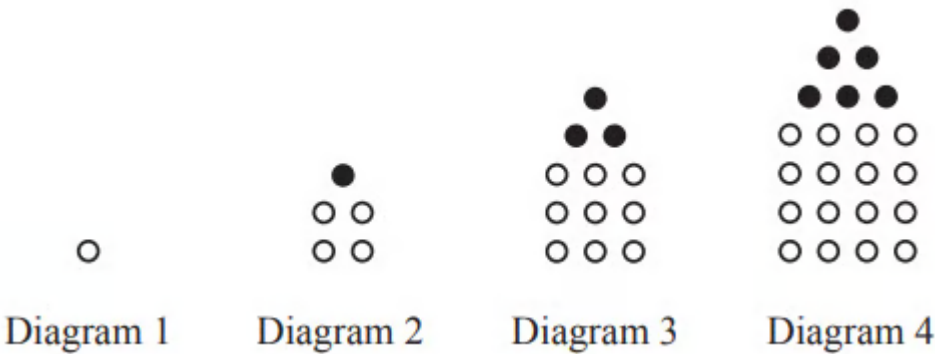
6 Find the n th term of the sequence below.

3, 6, 12, 24, ...

(2 marks)

Very Hard Questions

1 (a)



These are the first four diagrams of a sequence.
 The diagrams are made from white dots and black dots.

Complete the table for Diagram 5 and Diagram 6.

Diagram	1	2	3	4	5	6
Number of white dots	1	4	9	16		
Number of black dots	0	1	3	6		
Total number of dots	1	5	12	22		

(2 marks)

(b) Write an expression, in terms of n , for the number of white dots in Diagram n .

(1 mark)

(c) The expression for the total number of dots in Diagram n is $\frac{1}{2}(3n^2 - n)$.

i) Find the total number of dots in Diagram 8.

[1]

ii) Find an expression for the number of black dots in Diagram n . Give your answer in its simplest form.

[2]

(3 marks)

(d) T is the total number of dots used to make **all** of the first n diagrams.

$$T = an^3 + bn^2$$

Find the value of a and the value of b . You must show all your working.

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(5 marks)

2 The table shows the first four terms in sequences A , B , and C .

Sequence	1st term	2nd term	3rd term	4th term	5th term		n th term
A	4	9	14	19			
B	3	10	29	66			
C	1	4	16	64			

Complete the table.

(9 marks)

3 (a) The table shows the first five terms of sequences A , B and C .

Sequence	1st term	2nd term	3rd term	4th term	5th term	6th term
A	0	1	4	9	16	
B	4	5	6	7	8	
C	-4	-4	-2	2	8	

Complete the table.

(3 marks)

(b) Find an expression for the n th term of

i) sequence A ,

[2]

ii) sequence B .

[1]

(3 marks)

(c) Find the value of n when the n th term of sequence A is 576.

$n = \dots\dots\dots$

(2 marks)

(d) i) Find an expression for the n th term of sequence C. Give your answer in its simplest form.

[3]

ii) Find the value of the 30th term of sequence C.

[2]

(5 marks)